

Patrick Maxiao Ma

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Education

University of Waterloo, Waterloo, ON
Bachelor of Computer Science (Co-op Program)
Cumulative Average: 92

Sep. 2024 – Apr. 2029

Work Experience

Software Engineer Intern – AI Agent Team

<https://www.jurin.ai/>

May. 2026 – Aug. 2026

Jurin AI, Tokyo, Japan

- Worked on a production **AI-powered enterprise workspace**, building full-stack product features and automation workflows using **TypeScript, tRPC, Prisma/PostgreSQL, and AWS**.
- Built identity-resolution workflows for inbound calls and WhatsApp, including **Meta API** integrations and backward-compatible handling of evolving WhatsApp identifiers.
- Implemented **role-based authorization** across workspace agents and chats, updating application access logic and migrating production data as the permission model evolved.
- Implemented and extended **agent tools** for a production voice platform handling **10,000+ calls/month**, including language detection and human escalation with **LiveKit/Twilio** handoff logic.
- Contributed to **RAG and knowledge-prefetch systems**, building full-stack support for retrieval parameters, context limits, knowledge selection, and related production data migrations.

Full Stack Developer Intern

<https://www.sparklease.com/>

May. 2025 – Aug. 2025

SparkLease, Toronto, Canada

- Developed production backend services for an automotive marketplace serving **100,000+ annual users** using **ASP.NET Core MVC, Azure, and C#**.
- Implemented server-side **request deduplication and caching** with **Azure Blob Storage and SQL Server**, reducing network load by **50%+** and query latency by **56%**.
- Built multithreaded **Azure WebJobs** for API data ingestion and preprocessing, reducing processing time by **83%**.

Machine Learning Intern

<https://www.naxi-tech.com/>

Jun. 2024 – Aug 2024

NanoInsights, Beijing, China

- Trained and iteratively tuned **GAN-based super-resolution models** in **PyTorch** for **2× electron microscopy image enhancement**, improving PSNR from **~41 dB to ~45 dB**.
- Built an end-to-end data collection and **training pipeline** in **Python** by developing **crawling, filtering, and batching** workflows to ingest large-scale electron microscopy datasets and feed them into automated model training.
- Developed image preprocessing pipelines in **Python** using **NumPy, OpenCV, and scikit-image** to identify and filter low-information regions, expanding the usable training dataset by **80×** through geometric augmentation.

Projects

SEO AI Agent 📄

- Developed a full-stack SEO optimization agent using **FastAPI, Gemini API, and Docker**, generating HTML fixes that raised Lighthouse SEO scores to 90+ in under **30 seconds**.
- Designed a structured **agent pipeline** that orchestrates data fetching, analysis, patch generation, and validation.

LACS Compiler 📄

- Built an end-to-end compiler for **LACS, a Scala-like imperative language**, targeting **MIPS machine code**.
- Implemented a compiler pipeline with **scanning, parsing, type checking, code generation, and IR transformations**, including **closure support, tail-call optimization, and a semispace copying garbage collector**.

ASCII Game Engine

- Implemented a **C++ game engine** with **MVC architecture**, reusable **design patterns**, and a **physics engine**.
- Built a **reinforcement learning agent** that autonomously plays a **Mario-like game** developed based on the engine.

PianoFi 📄

- Refactored **FastAPI backend**, redesigning APIs and consolidating transcription workflows into a cleaner architecture.
- Contributed to backend integration, **Docker/AWS** deployment, and model training for an AI transcription platform.

Skills

- **Programming Languages:** Python, TypeScript, JavaScript, C++, C#, Java, C, Scala, Racket
- **Web & Backend:** React, Next.js, Node.js, tRPC, Prisma, FastAPI, ASP.NET Core
- **AI & ML:** PyTorch, TensorFlow, RAG, tool calling, AI agents, reinforcement learning
- **Infrastructure & Data:** AWS, Azure, Docker, PostgreSQL, LiveKit, Twilio, Linux